USB_CHARGER

USB-C input, BQ25895 charger / power-path, 1S battery

USB-C sink (Rd 5k1), TVS1400, TPD2E2U06

BQ25895: ILIM 2.37 A ceiling, OTG disabled in hardware

TS 103AT window, 8 A fuse, KK-396 battery

File: USB_Charger.kicad_sch

PVDD_BOOST

TPS61088-Q1 amplifier supply boost: VSYS (3.0-4.4 V) -> PVDD 9.3 V

VSYS -> 9.30 V, ~550 kHz, 2.2 uH XAL7030

ILIM 8.1 A typ, EN from ESP32 (pull-down)

PFM default, FPWM option R205

File: PVDD_Boost.kicad_sch

3V3_BUCKBOOST

ISL91127IR buck-boost: VSYS (2.5-4.4 V) -> +3V3 switched digital rail

VSYS -> 3.33 V, 1 uH XFL4020

EN_3V3 from ATtiny412 (pull-down)

Supplies ESP32, TAS5806M DVDD

File: 3V3_BuckBoost.kicad_sch

POWER_CTRL

Always-on domain: TPS7A2033 LDO, ATtiny412 power latch, encoder, UPDI

TPS7A2033 always-on 3.3 V (6.5 uA)

ATtiny412 latch: button, CHG_INT wake, UART to ESP32

EC11E encoder (A/B to ESP32), UPDI header

File: Power_Control.kicad_sch

ESP32

ESP32-WROOM-32UE: Bluetooth A2DP sink, DSP, I2S master, I2C host

ESP32-WROOM-32UE-N8R2, U.FL antenna

I2S master (22R), I2C host, CC sense ADC

RESET/BOOT, 6-pin programming header

File: ESP32.kicad_sch

AMPLIFIER

TAS5806M stereo Class-D (BTL), LC output filters, speaker outputs

TAS5806M BTL x2, PVDD 9.3 V

10 uH + 0.68 uF per leg (CSAD0660-100M)

Speaker L/R connectors, I2S test points

File: Amplifier.kicad_sch

Rails: VBUS 5 V (<=12 V) | VSYS 3.0-4.4 V (BQ25895 SYS) | VBAT | PVDD 9.3 V (boost) | +3V3 switched (ISL91127) | +3V3_AON (TPS7A2033)

Single GND everywhere (one plane). Inter-sheet signals are global labels; rails are power symbols.

Status: schematic generated and ERC-clean; netlist verified against design intent. NOT simulated, NOT bench tested.

Engineering notes: DESIGN_NOTES.md in this folder

Rev A schematic - generated 2026-10-19 - NOT yet reviewed on the bench, PCB not laid out

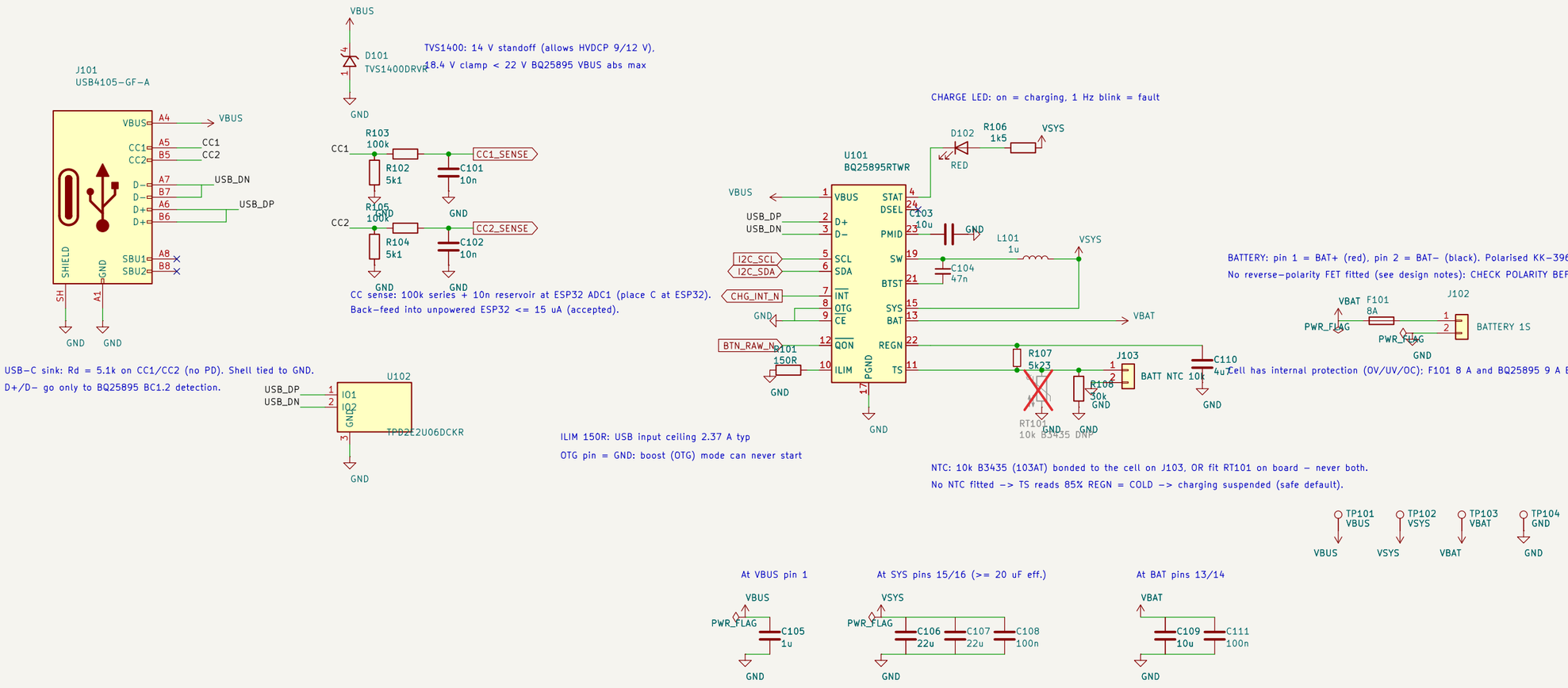
TAS5806M Bluetooth soundbar

Sheet: /

File: TAS5806M.kicad_sch

Title: 1S Bluetooth stereo soundbar (TAS5806M)

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/7

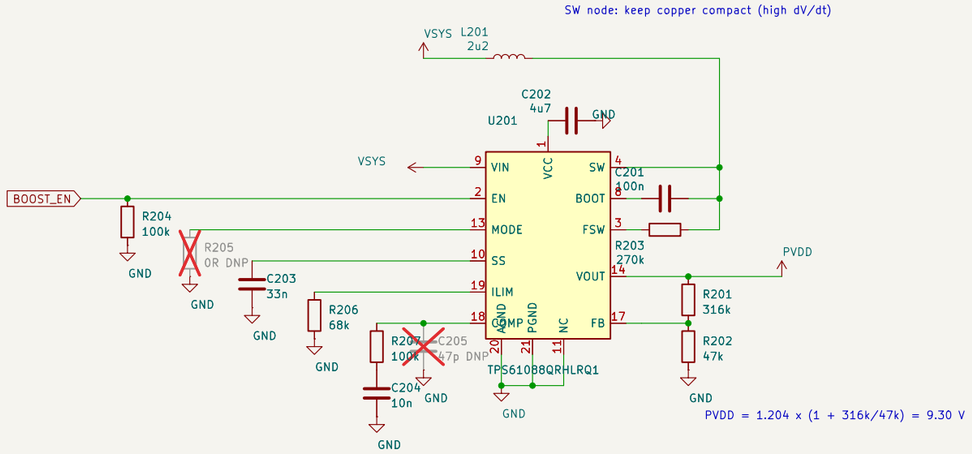


Battery: 9 Ah 1S pouch with internal protection; 8 A fuse in BAT+; KK-396 polarised connector
TS window with 103AT NTC: start 1-40 C, cut-off 44.5 C; NTC absent = charging inhibited
BQ25895: OTG pin hard-tied low (OTG_CONFIG defaults to 1), /CE tied low, ILIM ceiling 2.37 A
TAS5806M Bluetooth soundbar

Sheet: /USB_CHARGER/
File: USB_Charger.kicad_sch

Title: USB-C input, BQ25895 charger / power-path, 1S battery

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/7



MODE: open = PFM light-load (default). Fit R205 = forced PWM -> then change R206 to 56k.

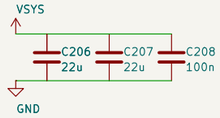
ILIM (PFM): 550000/68k = 8.1 A typ, >= 6.6 A worst case; inductor Isat 18 A.

Compensation: R207/C204 zero ~160 Hz, crossover ~7–12 kHz; C205 DNP (tuning only).

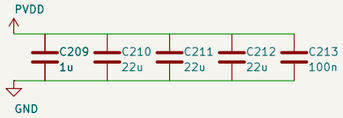
Soft start 33 nF -> 7.9 ms.



Boost input, at VIN/PGND (25 V X5R)



Boost output, at VOUT/PGND (25 V X5R; ~10 uF each at 9.3 V)



Loop: R5 100k / C5 10n, C8 DNP. Verify with a load-step test before connecting speakers

EN low = switching off; PVDD then sits at VSYS − body diode (TPS61088 has no output disconnect)

VOUT = 1.204 V x (1 + 316k/47k) = 9.30 V; fsw ~550 kHz (RFREQ 270k); ILIM 8.1 A typ (68k, PFM)

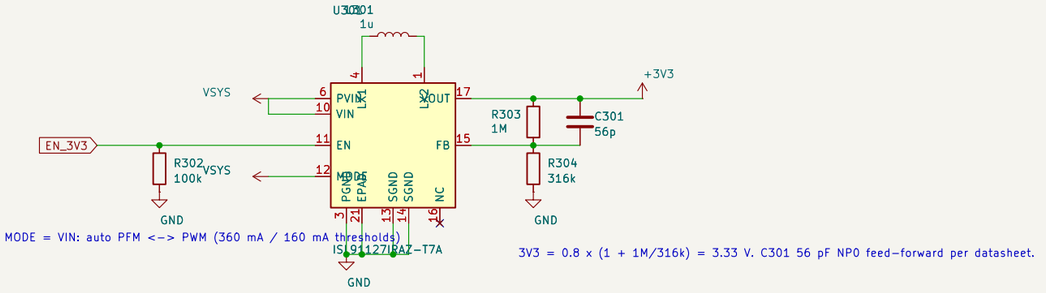
TAS5806M Bluetooth soundbar

Sheet: /PVDD_BOOST/

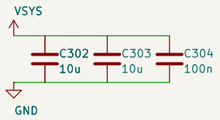
File: PVDD_Boost.kicad_sch

Title: TPS61088-Q1 amplifier supply boost: VSYS (3.0–4.4 V) -> PVDD 9.3

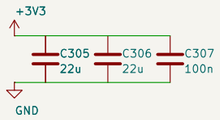
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/7



At PVIN/PGND (datasheet: 2 x 10 uF)



At VOUT/PGND (datasheet: 2 x 22 uF)





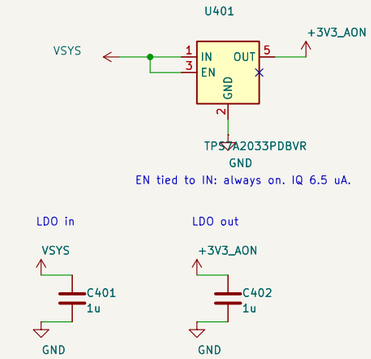
Loads: ESP32–WROOM–32UE (500 mA RF bursts), TAS5806M DVDD, I2C/I2S pull–ups, status LED
EN_3V3 from ATtiny412 (always–on domain); R302 holds the rail OFF when the ATtiny is unprogramme
VOUT = 0.8 V x (1 + 1M/316k) = 3.33 V; 2.5 MHz; auto PFM/PWM (MODE high)

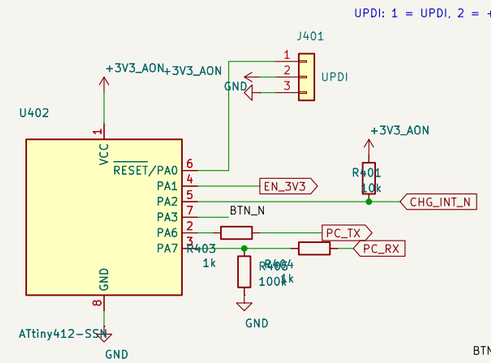
TAS5806M Bluetooth soundbar

Sheet: /3V3_BUCKBOOST/
File: 3V3_BuckBoost.kicad_sch

Title: ISL91127IR buck–boost: VSYS (2.5–4.4 V) –> +3V3 switched digital r

Size: A3	Date: 2026–10–19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/7

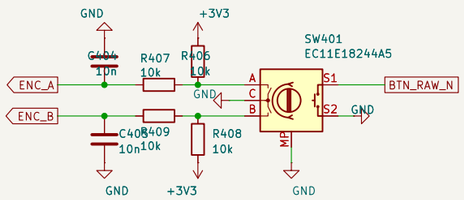












Encoder A/B: 10k pull-up + 10k/10n RC per encoder datasheet practice; pull-ups on switched +3V3 (no back-feed when the ESP32 is off). Contacts to C = GND. Push switch S1-S2 to GND.

Encoder bushing/MP tabs to GND for ESD.

ATtiny pins: PA0 UPDI, PA1 EN_3V3, PA2 CHG_INT_N (async wake), PA3 BTN_N, PA6 TXD, PA7 RXD
Short press = ON, hold >= 2 s = graceful OFF via ESP32, hold >= 12 s (no USB) = BQ25895 /QON
OFF state: only BQ25895 (BATFET on), TPS7A2033 and sleeping ATtiny412 draw current (~40 uA typ)

TAS5806M Bluetooth soundbar

Sheet: /POWER_CTRL/
File: Power_Control.kicad_sch

Title: Always-on domain: TPS7A2033 LDO, ATtiny412 power latch, encoder, U

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/7

U501
ESP32-WROOM-32UE-N8R2

GND
+3V3
ESP_EN
CC1_SENSE
CC2_SENSE
ENC_A
ENC_B
PC_TX
PC_RX
I2S_LRCLK
I2S_BCLK
I2S_DOUT
R502 22R
R503 22R
R504 22R
TP501 IO14
GND
D501 GREEN
R505 1k
GND
GND_1 1
3V3 2
EN 3
SENSOR_VP 4
SENSOR_VN 5
IO34 6
IO35 7
IO32 8
IO33 9
IO25 10
IO26 11
IO27 12
IO14 13
IO12 14
GND_2 15
IO13 16
NC_1 17
NC_2 18
NC_3 19
NC_4 20
NC_5 21
NC_6 22
IO15 23
IO2 24
IO0 25
IO4 26
IO16 27
IO17 28
IO5 29
IO18 30
IO19 31
NC_7 32
IO21 33
RXD0 34
TXD0 35
IO22 36
IO23 37
GND_3 38
GND_4 39
GND_5 40
GND_6 41
GND_7 42
GND_8 43
GND_9 44
GND_10 45
GND_11 46
GND_12 47
ESP_IO0
BOOST_EN
ESP_IO16
TP502 IO17
AMP_FAULT_N
TP503 IO19
I2C_SDA
ESP_RXD0
ESP_TXD0
I2C_SCL
AMP_PDN
GND

22R source terminations at the ESP32 pins (I2S lines to TAS5806M)

17–22: module NC. IO15/IO2/IO12: strapping, unused.

At module pin 2 (Espressif: 22 uF + 100 nF; +10 uF for RF bursts)

+3V3
C502 22u
C503 10u
C504 100n
GND

ANTENNA: WROOM-32UE uses an external U.FL antenna — route the coax away from PVDD/boost/Class-D outputs; no PCB keep-out is required for the UE variant (datasheet sec. 3.1).

EN RC 10k / 1u per Espressif (power-up delay)

+3V3
R501 10k
ESP_EN
SW501 RESET
1 3
C501 1u
GND
GND

SW502 BOOT
ESP_IO0
1 3
GND

IO0 has an internal pull-up; hold BOOT while pulsing RESET for the ROM loader.

+3V3
R506 10k
ESP_IO16

IO16 = PSRAM CS on N8R2: 10k pull-up (Espressif)

I2C pull-ups 4k7 to +3V3 (400 kHz, BQ25895 + TAS5806M).
Bus is unpowered (0 V) when +3V3 is off — BQ25895 SCL/SDA tolerate this.

+3V3
R507 4k7
I2C_SDA
+3V3
R508 4k7
I2C_SCL

J501
ESP PROG
1 +3V3
2 GND
3 ESP_TXD0
4 ESP_RXD0
5 ESP_EN
6 ESP_IO0

J501: 1 GND, 2 +3V3, 3 TXD0 (ESP out), 4 RXD0 (ESP in), 5 EN, 6 IO0.
Pin 2 may power the 3V3 rail from a 3.3 V adapter for bench flashing with the battery disconnected (ISL91127 off: its 120R output discharge draws ~27 mA — acceptable).

N8R2 module: IO16 is the PSRAM chip-select -> 10k pull-up only, never used as GPIO
Strapping pins IO0/IO2/IO5/IO12/IO15 left at defaults; IO12 must be low at boot (3.3 V flash)
I2S0 master 48 kHz, 32-bit slots: BCLK 3.072 MHz (64 fs), no MCLK (TAS5806M uses BCLK PLL)

TAS5806M Bluetooth soundbar

Sheet: /ESP32/
File: ESP32.kicad_sch

Title: ESP32-WROOM-32UE: Bluetooth A2DP sink, DSP, I2S master, I2C host

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 6/7

VR_DIG / AVDD: internal regulators — caps only, no external loads

U601
TAS5806MDCPR

PVDD
PVDD

+3V3

C614
1u
GND

C615
1u
GND

C601
220n
C602
220n

BST_A−
OUT_A−

AMP_PDN
AMP_FAULT_N
I2C_SCL
I2C_SDA
I2S_BCLK
I2S_LRCLK
I2S_DOUT

SDOUT (26) unused

C607
220n
C608
220n

BST_B+
OUT_B+
BST_B−
OUT_B−

GND

LEFT channel: SPK_L+ = pin 1, SPK_L− = pin 2 (BTL — do not ground either terminal)

CSAD0660−100M
L601

C603
680n
C605
1n

SPK_L+
SPK_L−

J601
SPEAKER L

C604
680n
C606
1n

GND
GND

RIGHT channel: SPK_R+ = pin 1, SPK_R− = pin 2 (BTL — do not ground either terminal)

CSAD0660−100M
L602

C609
680n
C611
1n

SPK_R+
SPK_R−

J602
SPEAKER R

C610
680n
C612
1n

GND
GND

At PVDD pins 6/7 (TI: 22 uF + 0.1 uF)

PVDD
C616
22u
C617
100n
GND

At PVDD pins 32/33

PVDD
C618
22u
C619
100n
GND

PVDD bulk 220u/16V FR

PVDD
C620
220u
GND

At DVDD pin 9 (TI: 4.7 uF + 0.1 uF)

+3V3
C621
4u7
C622
100n
GND

R601 15k to DVDD: I2C address 0x2D (8−bit 0x5A); pin re−defined as FAULT after init

+3V3
R601
15k
AMP_FAULT_N

AMP_PDN
R602
100k
GND

R602 holds PDN low (shutdown) while the ESP32 is off/booting

TP601
BCLK
I2S_BCLK

TP602
LRCLK
I2S_LRCLK

TP603
DOUT
I2S_DOUT

BTL outputs: NEITHER speaker terminal is ground. Bootstraps 0.22 uF per TI 10.1.1
Filter per leg: 10 uH (Codaca CSAD0660−100M winding) + 0.68 uF to GND −> 61 kHz (TI SLASEP9 Ta
PVDD 9.3 V, hybrid modulation, Fsw 384 kHz (or 768 kHz 1SPW), AGAIN −10.5 dB, I2C address 0x2D (

TAS5806M Bluetooth soundbar

Sheet: /AMPLIFIER/
File: Amplifier.kicad_sch

Title: TAS5806M stereo Class−D (BTL), LC output filters, speaker outputs

Size: A3	Date: 2026−10−19	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 7/7